Pizza Sales Analysis

Answering 13 business questions with SQL

Tool: MySQL Workbench

Skills: JOINs, Aggregations, Subqueries, Window Functions



OVERVIEW

Project Overview

What this project does

- Analyses a year of pizza-restaurant orders to understand sales, customer preferences and peak hours.
- Questions are grouped by difficulty: **5 Basic**, **5 Intermediate**, **3 Advanced**.
- Advanced queries use subqueries, CTE-style derived tables and window functions (SUM OVER, RANK OVER).

Database schema

orders
order_id
order_date
order_time

orders_details
order_details_id
order_id
pizza_id
quantity

pizzas
pizza_id
pizza_type_id
size
price

pizza_types
pizza_type_id
name
category
ingredients

BASIC

Question 1

Retrieve the total number of orders placed.

```
15 •  SELECT
16        COUNT(order_id) AS Total_orders
17    FROM
18        orders;
19
20    Calculate the total revenue generated from pizza sales
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_orders
21350

KEY INSIGHT

21,350 orders were placed in total.

BASIC

Question 2

Calculate the total revenue generated from pizza sales.

```
21 •  SELECT
22        ROUND(SUM(orders_details.quantity * pizzas.price),
23                2) AS Total_sales
24    FROM
25        orders_details
26            JOIN
27        pizzas ON pizzas.pizza_id = orders_details.pizza_id;
28
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Total_sales
817860.05

KEY INSIGHT

Pizza sales generated $817,860.05 in revenue.

BASIC

Question 3

Identify the highest-priced pizza.

```sql
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY price DESC
LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	price
The Greek Pizza	35.95

KEY INSIGHT

The Greek Pizza is the most expensive item on the menu at $35.95.

BASIC

Question 4

Identify the most common pizza size ordered.

```
40 •  SELECT
41        pizzas.size,
42        COUNT(orders_details.order_details_id) AS order_count
43    FROM
44        pizzas
45            JOIN
46        orders_details ON pizzas.pizza_id = orders_details.pizza_id
47    GROUP BY pizzas.size
48    ORDER BY order_count DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

size	order_count
L	18526
M	15385
S	14137
XL	544
XXL	28

KEY INSIGHT

Large (L) is the customer favourite with 18,526 order lines, ahead of M (15,385) and S (14,137). XL and XXL are rare.

BASIC

Question 5

List the top 5 most ordered pizza types along with their quantities.

```sql
SELECT
    pizza_types.name, SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	quantity
The Classic Deluxe Pizza	2453
The Barbecue Chicken Pizza	2432
The Hawaiian Pizza	2422
The Pepperoni Pizza	2418
The Thai Chicken Pizza	2371

KEY INSIGHT

Classic Deluxe leads with 2,453 pizzas - but the top 5 are separated by only ~80 units.

INTERMEDIATE

Question 6

Join the necessary tables to find the total quantity of each pizza category ordered.

```sql
SELECT
    pizza_types.category,
    SUM(orders_details.quantity) AS quantity
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	quantity
Classic	14888
Supreme	11987
Veggie	11649
Chicken	11050

KEY INSIGHT

Classic is the most ordered category (14,888 pizzas), followed by Supreme (11,987), Veggie (11,649) and Chicken (11,050).

INTERMEDIATE

Question 7

Determine the distribution of orders by hour of the day.

```
78 •  SELECT
79        HOUR(order_time) AS HOUR, COUNT(order_id) AS order_count
80    FROM
81        orders
82    GROUP BY HOUR(order_time);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

HOUR	order_count
11	1231
12	2520
13	2455
14	1472
15	1468
16	1920
17	2336
18	2399

KEY INSIGHT

Two clear rushes: lunch (12:00-13:00, ~2,500 orders per hour) and dinner (17:00-18:00, ~2,300-2,400 per hour).

INTERMEDIATE

Question 8

Join relevant tables to find the category-wise distribution of pizzas.

```
85 •  SELECT
86        category, COUNT(name)
87    FROM
88        pizza_types
89    GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	COUNT(name)
Chicken	6
Classic	8
Supreme	9
Veggie	9

KEY INSIGHT

The menu has 32 pizza types: Supreme (9), Veggie (9), Classic (8) and Chicken (6).

INTERMEDIATE

Question 9

Group the orders by date and calculate the average number of pizzas ordered per day.

```sql
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(orders_details.quantity) AS quantity
    FROM
        orders
    JOIN orders_details ON orders.order_id = orders_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

ROUND(AVG(quantity),0)
138

KEY INSIGHT

On average, about 138 pizzas are sold per day.

INTERMEDIATE

Question 10

Determine the top 3 most ordered pizza types based on revenue.

```sql
SELECT
    pizza_types.name,
    SUM(orders_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5

KEY INSIGHT

Chicken pizzas dominate revenue: Thai Chicken ($43,434), Barbecue Chicken ($42,768) and California Chicken ($41,410).

ADVANCED

Question 11

Calculate the percentage contribution of each pizza category to total revenue.

```sql
SELECT
    pizza_types.category,
    ROUND(SUM(orders_details.quantity * pizzas.price) / (SELECT
                    ROUND(SUM(orders_details.quantity * pizzas.price),
                                2) AS total_sales
                FROM
                    orders_details
                        JOIN
                    pizzas ON pizzas.pizza_id = orders_details.pizza_id) * 100,
            2) AS revenue
FROM
    pizza_types
        JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN
    orders_details ON orders_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	revenue
Classic	26.91
Supreme	25.46
Chicken	23.96
Veggie	23.68

KEY INSIGHT

Revenue is spread evenly: Classic 26.91%, Supreme 25.46%, Chicken 23.96%, Veggie 23.68%.

ADVANCED

Question 12

Analyze the cumulative revenue generated over time.

```
137 •   SELECT order_date,
138     SUM(revenue) OVER (ORDER BY order_date) AS cum_revenue
139     FROM (SELECT orders.order_date, SUM(orders_details.quantity*pizzas.price) AS revenue
140     FROM orders_details JOIN pizzas
141     ON orders_details.pizza_id = pizzas.pizza_id
142     JOIN orders
143     ON orders.order_id = orders_details.order_id
144     GROUP BY orders.order_date) AS sales
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

order_date	cum_revenue
2015-01-01	2713.8500000000004
2015-01-02	5445.75
2015-01-03	8108.15
2015-01-04	9863.6
2015-01-05	11929.55
2015-01-06	14358.5
2015-01-07	16560.7
2015-01-08	19399.05

KEY INSIGHT

A window function builds a running total - about $19.4K in revenue within the first 8 days of 2015.

ADVANCED

Question 13

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
147 •  SELECT name, revenue FROM
148    (SELECT category, name, revenue,
149    RANK() OVER ( partition by category ORDER BY revenue DESC) AS rn
150    FROM
151    (SELECT pizza_types.category, pizza_types.name,
152    SUM((orders_details.quantity) * pizzas.price) AS revenue
153    FROM pizza_types JOIN pizzas
154    ON pizza_types.pizza_type_id = pizzas.pizza_type_id
155    JOIN orders_details
156    ON orders_details.pizza_id = pizzas.pizza_id
157    GROUP BY pizza_types.category, pizza_types.name) AS a) AS b
158    WHERE rn<= 3
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	revenue
The Thai Chicken Pizza	43434.25
The Barbecue Chicken Pizza	42768
The California Chicken Pizza	41409.5
The Classic Deluxe Pizza	38180.5
The Hawaiian Pizza	32273.25
The Pepperoni Pizza	30161.75
The Spicy Italian Pizza	34831.25
The Italian Supreme Pizza	33476.75

KEY INSIGHT

RANK() partitioned by category keeps the 3 best-earning pizzas in each category.

SUMMARY

Key Takeaways

$817.9K

total revenue from 21,350 orders

138

pizzas sold per day on average

12-13 & 17-18

lunch and dinner peak hours

Large

most popular pizza size

Classic

top category by volume (26.9% of revenue)

Thai Chicken

highest-earning pizza ($43.4K)